
NHSNLink Patient List Creation Instructions (Epic)

Background

Epic exposes two types of patient “lists” to FHIR API requests: Patient Lists (system or personal) and registries (via Bulk Data Request API).

NHSNLink uses the STU3/List API to retrieve patient IDs from system patient lists, so that admitted and recently admitted patient lists can be queried daily. This provides a historical, complete list of patient IDs for admitted patients from the prior month. NHSNLink retrieves lists from the following URL, which uses the [List.Read \(Patient List\) \(STU3\)](#) API:

/api/FHIR/STU3/List/{ID}

This API has a limit of 1500 patients returned with any one query, so we recommend creating the six system lists in Table 1 to keep daily queries well under this 1500-patient threshold and to avoid missing patients with short admissions. You can and should hide these lists from non-administrative users.

Run the [SQL Query Example](#) to determine the sum number of patients expected by historical date for the group of the first three lists and the group of the last three lists. If either of these values is greater than 1500 when completing that query for a 6–12-month period, please contact NHSNLink for instructions before completing these build recommendations.

NHSNLink requires the internal ID (PAC .1) for the system lists in the table on the following page. Build instructions follow the table.

Record PAC ID here	System List Grouper	List Name	Intent/Description	Initial Number of Patients in SUP
	NHSNLink Admitted Patients	Admit w/in 24h	Currently admitted, admission start within past 24 hours	
	NHSNLink Admitted Patients	Admit b/t 24-48h	Currently admitted, admission start between 24 and 48 hours ago	
	NHSNLink Admitted Patients	Admit over 48h	Currently admitted, admission start more than 48 hours ago	
	NHSNLink Recently Discharged	Admit w/in 24h	Recently discharged, lookback 2 days, admission start within past 24 hours	
	NHSNLink Recently Discharged	Admit b/t 24-48h	Recently discharged, lookback 2 days, admission start between 24 and 48 hours ago	
	NHSNLink Recently Discharged	Admit over 48h	Recently discharged, lookback 2 days, admission start more than 48 hours ago	

Table 1: Patient Lists and IDs for NHSNLink

Build Overview

Rule Creation

Create a rule that represents admission time within the ranges specified above.

1. In Hyperspace Rule Editor, create a rule with the following attributes:
 - a. Context: **Patient [2001]**
 - b. Name: **<CUSTOMER PREFIX/NAMING CONVENTION> Patient admission start within the last 24 hours**
 - c. Record ID: _____
Default: **AND**
 - d. Line 1
 - i. Property: **Patient >> Hospital Admission Instant**
 - ii. Operator: **>**
 - iii. Value: **Constant >> FTime in Past or Future**
 1. Minutes + or - : **-1440**
 2. UTC?: **<NULL>**
 - e. Line 2
 - i. Property: **Patient >> ADT Patient Class**
 - ii. Operator: **=**
 - iii. Values:
 1. **<all values that represent an *inpatient admission** at the facility**>**
 2. **<all values that represent an *ED visit** at the facility**>**
 3. **<all values that represent an *observation** at the facility**>**
 4. **<all values that represent a *short stay** at the facility**>**

**These should include all patients of all ages, including newborns.*

***For hospital systems that report using more than 1 NHSN Facility/Org ID, facility equals one specific NHSN Facility/Org ID*

NHSNCoLab facilities will work directly with the NHSNCoLab team to identify which ADT Patient Class codes should be included in the initial list build.

- iv. Error message: **"Admission start within last 24 hours"**
2. Save rule (not Accept).
3. Duplicate that rule (Open >> Create a new rule >> Copy from:, etc.) into a new rule with the following attributes:
 - a. Context: **Patient [2001]**

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- b. Name: **<CUSTOMER PREFIX/NAMING CONVENTION> <CUSTOMER PREFIX/NAMING CONVENTION>Patient admission start between 24 hours and 48 hours ago**
 - c. Record ID: _____
 - d. Evaluation logic: **And**
 - e. Line 1
 - i. Property: **Patient >> Hospital Admission Instant**
 - ii. CHANGE THIS: Operator: **<=**
 - iii. Value: **Constant >> Time in Past or Future**
 - 1. Minutes + or - : **-1440**
 - 2. UTC?: **<NULL>**
 - f. Confirm Line 2
 - i. Property: **Patient >> ADT Patient Class**
 - ii. Operator: **=**
 - iii. Values:
 - 1. **<all values that represent an *inpatient admission** at the facility**>**
 - 2. **<all values that represent an *ED visit** at the facility**>**
 - 3. **<all values that represent an *observation** at the facility**>**
 - 4. **<all values that represent a *short stay** at the facility**>**

**These should include all patients of all ages, including newborns.*

***For hospital systems that report using more than 1 NHSN Facility/Org ID, facility equals one specific NHSN Facility/Org ID*

NHSNCoLab facilities will work directly with the NHSNCoLab team to identify which ADT Patient Class codes should be included in the initial list build.

- g. Add Line 3
 - i. Property: **Patient >> Hospital Admission Instant**
 - ii. Operator: **>**
 - iii. Value: **Constant >> Time in Past or Future**
 - 1. Minutes + or - : **-2880**
 - 2. UTC?: **<NULL>**
 - h. CHANGE THIS: Error message: **"Admission start between 24 and 48 hours ago"**
 - 4. Accept rule.
 - 5. Duplicate the first rule again (Open >> Create a new rule >> Copy from:, etc.) into a new rule with the following attributes:
 - a. Context: **Patient [2001]**
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- b. Name: **<CUSTOMER PREFIX/NAMING CONVENTION> Patient admission start over 48 hours ago**
 - c. Record ID: _____
 - d. Line 1
 - i. Property: **Patient >> Hospital Admission Instant**
 - ii. CHANGE THIS: Operator: **<=**
 - iii. Value: **Constant >> Time in Past or Future**
 - 1. Minutes + or - : **-2880**
 - 2. UTC?: **<NULL>**
 - e. Confirm Line 2
 - i. Property: **Patient >> ADT Patient Class**
 - ii. Operator: **=**
 - iii. Values:
 - 1. **<all values that represent an *inpatient admission** at the facility**>**
 - 2. **<all values that represent an *ED visit** at the facility**>**
 - 3. **<all values that represent an *observation** at the facility**>**
 - 4. **<all values that represent a *short stay** at the facility**>**
- *These should include all patients of all ages, including newborns.*
- **For hospital systems that report using more than 1 NHSN Facility/Org ID, facility equals one specific NHSN Facility/Org ID*
- NHSNCoLab facilities will work directly with the NHSNCoLab team to identify which ADT Patient Class codes should be included in the initial list build.*
- iv. CHANGE THIS: Error message: **"Admission start over 48 hours ago"**
6. Accept and close all rules.

Extension Creation

Create columns that display the error messages for the rules, to be used as a filter for the patient lists. These first require extensions.

- 7. In Text, go to Chronicles: LPP (Extensions).
 - 8. Enter Data >> Duplicate Extension
 - 9. Duplicate 74459 into a new Extension with the following properties:
 - a. Record ID: _____
 - b. Extension name: **<CUSTOMER PREFIX/NAMING CONVENTION> Patient admission start within last 24 hours**
 - 10. Edit extension and modify the first parameter to match the following:
 - a. 1. Rules: **Matching ID from Step 1.c above**
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11. Repeat this twice more to create extensions with the following properties/parameters:
 - a. Record ID: _____
 - i. Extension name: <CUSTOMER PREFIX/NAMING CONVENTION> **Patient admission start between 24 and 48 hours ago**
 - ii. Parameters
 1. Rules: **Matching ID from Step 3.c above**
 - b. Record ID: _____
 - i. Extension name: <CUSTOMER PREFIX/NAMING CONVENTION> **Patient admission start over 48 hours ago**
 - ii. Parameters
 1. Rules: **Matching ID from Step 5.c above**
 12. Close Extension editor in Chronicles.

Column Creation

Complete creating the columns.

13. In Column Editor, Create New Column
 - a. Copy from: **Rule Evaluation [2286]**
 - b. Column ID: _____
 - c. Column name: <CUSTOMER PREFIX/NAMING CONVENTION> **Patient admission start within last 24 hours**
 - d. Caption: CHANGE TO: **Admitted w/in 24 hours**
 - e. Used by: ADD: **Inpatient EMR [26]**
 - f. Field scope: CHANGE TO: **Filter Only**
 - g. Text Ext: CHANGE TO: **Record ID from Step 9.a above**
 - h. Accept column to save it.
 14. In Column Editor, Create New Column
 - a. Copy from: New column you just created, with ID in Step b above.
 - b. Column ID: _____
 - c. Column name: <CUSTOMER PREFIX/NAMING CONVENTION> **Patient admission start between 24 and 48 hours ago**
 - d. Caption: CHANGE TO: **Admitted b/t 24-48 hours ago**
 - e. Text Ext: CHANGE TO: **Record ID from Step 11.a above**
 - f. Accept column to save it.
 15. In Column Editor, Create New Column
 - a. Copy from: New column you just created, with ID in Step b above.
 - b. Column ID: _____
 - c. Column name: <CUSTOMER PREFIX/NAMING CONVENTION> **Patient admission start over 48 hours ago**
 - d. Caption: CHANGE TO: **Admitted over 48 hours ago**
 - e. Text Ext: CHANGE TO: **Record ID from Step 11.b above**
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- f. Accept column to save it.

Patient List Creation

Create Patient Lists, using the newly created columns as the filters.

16. In Patient Lists activity, find a System List Grouper (folder) within System Lists (one that represents All Admitted Patients is a good choice), right click the folder, click Properties, and record the Record ID in the header of the window here: _____.
17. Click on Cancel to exit the System List Grouper without making any changes.
18. Right click on the System Lists folder above the folder you just found and click Create System List Grouper.
19. Create a new System List Grouper to view filtered lists of currently admitted patients:
 - a. Name: **NHSNLink Admitted Patients**
 - b. Click Copy.
 - c. In “Or, copy from:”, type the ID you recorded in Step 16.
 - d. Click Accept.
 - e. In the Advanced tab, add the background user for your NHSNLink App to the Allowed Users field.
 - i. This background user will need to have one or more of the following security points ([List Resource Info on Galaxy](#)):
 1. EpicCare Ambulatory security point 109-Patient List Administrator
 2. EpicCare Inpatient security point 1-Patient List Administrator
 - f. Click Accept.
20. Scroll down and find the new System List Grouper.
21. Right-click on NHSNLink Admitted Patients and click Create System List.
22. Create a new System List with the following properties:
 - a. Name: **Admit w/in 24h**
 - b. Group: **NHSNLink Admitted Patients**
 - c. Patient source: **Hospital Census**
 - d. Include pended patients: **UNCHECKED**
 - e. Include patients on Leave of Absence: **CHECKED**
 - f. Bed Occupancy Status: **Occupied Beds Only**
 - g. Limit Facility to those that would include inpatients meant to be submitted to NHSNLink.
 - h. Go to the Filters tab.
 - i. Scroll down to find “**Error! Reference source not found.**”.

NOTE: If you do not see your column(s), there are two possible problems. You may not have added Inpatient EMR [26] in the Used By field for the column, OR you may not have made sure that the column type included “Filter Only.” If the type is “Displayable Field Only,” it will not appear here for selection.

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- j. Add criteria and EXCLUDE “**cases with empty return value**”.
 - k. If you are assigning an ID manually, record the ID in **Error! Reference source not found.** now.
 - l. Otherwise, Accept the list, right-click on the list, and select Properties, then transcribe the Patient List ID that has been assigned by the system to **Error! Reference source not found.**.
23. Create an additional System List with the following properties:
- a. Name: **Admit b/t 24-48h**
 - b. Group: **NHSNLink Admitted Patients**
 - c. Patient source: **Hospital Census**
 - d. Include pended patients: **UNCHECKED**
 - e. Include patients on Leave of Absence: **CHECKED**
 - f. Bed Occupancy Status: **Occupied Beds Only**
 - g. Limit Facility to those that would include inpatients meant to be submitted to NHSNLink, as in Step 22.g. **If the facility is not assigned, the list will not work.**
 - h. Go to the Filters tab.
 - i. Scroll down to find “**Error! Reference source not found.**”.
 - j. Add criteria and EXCLUDE “**cases with empty return value**”.
 - k. If you are assigning an ID manually, record the ID in **Error! Reference source not found.** now.
 - l. Otherwise, Accept the list, right-click on the list, and select Properties, then transcribe the Patient List ID that has been assigned by the system to **Error! Reference source not found.**.
24. Create an additional System List with the following properties:
- a. Name: **Admit over 48h**
 - b. Patient source: **Hospital Census**
 - c. Include pended patients: **UNCHECKED**
 - d. Include patients on Leave of Absence: **CHECKED**
 - e. Bed Occupancy Status: **Occupied Beds Only**
 - f. Limit Facility to those that would include inpatients meant to be submitted to NHSNLink, as in Step 22.g.
 - g. Go to the Filters tab.
 - h. Scroll down to find “**Error! Reference source not found.**”.
 - i. Add criteria and EXCLUDE “**cases with empty return value**”.
 - j. If you are assigning an ID manually, record the ID in **Error! Reference source not found.** now.
 - k. Otherwise, Accept the list, right-click on the list, and select Properties, then transcribe the Patient List ID that has been assigned by the system to **Error! Reference source not found.**.
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25. In Patient Lists activity, find a System List Grouper within the Recently Discharged folder. Right click on the System List Grouper (folder), click Properties, and record the Record ID in the header of the window here: _____.
 26. Click on Cancel to exit the System List Grouper without making any changes.
 27. Right click on the Recently Discharged folder and click Create System List Grouper.
 28. Create a new System List Grouper to view filtered lists of currently admitted patients:
 - a. Name: **NHSNLink Recently Discharged**
 - b. Click Copy.
 - c. In “Or, copy from:”, type the ID you recorded in Step 25.
 - d. Click Accept.
 - e. In the Advanced tab, add the background user to the Allowed Users field.
 - i. This background user will need to have one or more of the following security points ([List Resource Info on Galaxy](#)):
 1. EpicCare Ambulatory security point 109-Patient List Administrator
 2. EpicCare Inpatient security point 1-Patient List Administrator
 - f. Click Accept.
 29. Scroll down and find the new System List Grouper.
 30. Right-click on **NHSNLink Recently Discharged** and click Create System List.
 31. Create a new System List with the following properties:
 - a. Name: **Admit w/in 24h**
 - b. Group: NHSNLink Recently Discharged
 - c. Facility: Limit Facility to those that would include inpatients meant to be submitted to NHSNLink – this should be the same as in Step 22.g.
 - d. Days to look back: **2**
 - e. Status: **Discharged: CHECKED**
 - f. Status: **Hospital Outpatient Visit: UNCHECKED**
 - g. Go to the Filters tab.
 - h. Scroll down to find “**Error! Reference source not found.**”
 - i. Add criteria and EXCLUDE “**cases with empty return value**”.
 - j. If you are assigning an ID manually, record the ID in **Error! Reference source not found.** now.
 - k. Otherwise, Accept the list, right-click on the list, and select Properties, then transcribe the Patient List ID that has been assigned by the system to **Error! Reference source not found.**
 32. Create a new System List with the following properties:
 - a. Name: **Admit b/t 24-48h**
 - b. Group: NHSNLink Recently Discharged
 - c. Facility: Limit Facility to those that would include inpatients meant to be submitted to NHSNLink – this should be the same as in Step 22.g.
 - d. Days to look back: **2**
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- e. Status: **Discharged: CHECKED**
 - f. Status: **Hospital Outpatient Visit: UNCHECKED**
 - g. Go to the Filters tab.
 - h. Scroll down to find “**Error! Reference source not found.**”
 - i. Add criteria and EXCLUDE “**cases with empty return value**”.
 - j. If you are assigning an ID manually, record the ID in ***Error! Reference source not found.*** now.
 - k. Otherwise, Accept the list, right-click on the list, and select Properties, then transcribe the Patient List ID that has been assigned by the system to ***Error! Reference source not found.***
33. Create a new System List with the following properties:
- a. Name: **Admit over 48h**
 - b. Group: NHSNLink Recently Discharged
 - c. Facility: Limit Facility to those that would include inpatients meant to be submitted to NHSNLink – this should be the same as in Step 22.g.
 - d. Days to look back: **2**
 - e. Status: **Discharged: CHECKED**
 - f. Status: **Hospital Outpatient Visit: UNCHECKED**
 - g. Go to the Filters tab.
 - h. Scroll down to find “**Error! Reference source not found.**”
 - i. Add criteria and EXCLUDE “**cases with empty return value**”.
 - j. If you are assigning an ID manually, record the ID in ***Error! Reference source not found.*** now.
 - k. Otherwise, Accept the list, right-click on the list, and select Properties, then transcribe the Patient List ID that has been assigned by the system to ***Error! Reference source not found.***
34. Confirm ***Error! Reference source not found.*** has all six IDs entered.
35. Check the number of patients that fall on these lists as soon as possible in SUP or some other copy of production environment. Record this number in ***Error! Reference source not found.***
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SQL Query Example

The following is an SQL query to confirm that the counts of proposed lists will be under the 1500 threshold. It is written for Oracle SQL Developer, so you may need to change the syntax to work in your system. You need to modify the second highlighted area to represent your location(s). You can modify the first highlighted date to look at a larger number of dates.

```

with hosp_enc as (
SELECT
    PH.PAT_ENC_CSN_ID,
    PH.HOSP_ADMSN_TIME,
    PH.HOSP_DISCH_TIME,
    ZC_PC.NAME AS PT_CLASS
FROM PAT_ENC_HSP PH
    LEFT OUTER JOIN ZC_PAT_CLASS ZC_PC ON PH.ADT_PAT_CLASS_C = ZC_PC.ADT_PAT_CLASS_C
    LEFT OUTER JOIN CLARITY_DEP CD ON PH.DEPARTMENT_ID = CD.DEPARTMENT_ID
    LEFT OUTER JOIN CLARITY_LOC CL ON CD.REV_LOC_ID = CL.LOC_ID
WHERE
    PH.HOSP_DISCH_TIME >= '1-JAN-2022'
    AND ZC_PC.NAME IN ('Inpatient', 'Observation', 'Outpatient in a Bed', 'Obs Greater than 48 Hours',
'Extended Recovery')
    AND CL.LOC_NAME IN ('CARDIOVASCULAR CENTER', 'UNIVERSITY HOSPITAL', 'UNIVERSITY HOSPITAL
SOUTH')),

admit_summary as (
select
    calendar_dt,
    count(HA.pat_enc_csn_id) admitted_at_midnight
from date_dimension
    join hosp_enc HA on ha.hosp_admsn_time < date_dimension.calendar_dt and ha.hosp_disch_time >
date_dimension.calendar_dt
where
    calendar_dt > (select min(hosp_admsn_time) from hosp_enc)
    and calendar_dt < (select max(hosp_disch_time) from hosp_enc)
group by calendar_dt),

discharge_summary as (
select
    calendar_dt,
    count(HD.pat_enc_csn_id) discharged_in_last_two_days
from date_dimension
    join hosp_enc HD on hd.hosp_disch_time < (date_dimension.calendar_dt) and hd.hosp_disch_time >
(date_dimension.calendar_dt - 2)
where
    calendar_dt > (select min(hosp_admsn_time) from hosp_enc)
    and calendar_dt < (select max(hosp_disch_time) from hosp_enc)

```

```
group by calendar_dt),

summary_by_date as (
select
    admit_summary.calendar_dt,
    admit_summary.admitted_at_midnight,
    discharge_summary.discharged_in_last_two_days
from admit_summary
    join discharge_summary on admit_summary.calendar_dt = discharge_summary.calendar_dt
order by calendar_dt desc)

select
    max(admitted_at_midnight),
    max(discharged_in_last_two_days)
from summary_by_date;
```